MAT2040

Tutorial 1&2

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About me and this course I

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- My learning experience of this course

Quick Review I

- The first part of linear algebra consider the following matrix representation of linear system

$$AX = b$$

  we have coefficent matrix A, constant vector b and argumented matrix $[A|b]$.It is homogeneous if b=0, otherwise it is non-homogeneous.

- Natural questions:

  When does the solution has a solution?How to find solution?

  How to determine whether 2 linear systems have same solution or not?

- Use REF and RREF!

Exercise

$$
\begin{aligned}
x_1 + x_2 \qquad\quad &+ 2x_5 = 3 \\
x_3 + x_4 + 5x_5 &= 4 \\
2x_1 + 2x_2 \quad + x_4 &+ 5x_5 = 9 \\
-3x_1 - 3x_2 + x_3 + 3x_4 &+ x_5 = 1
\end{aligned}
$$

(a) Write down the coefficient matrix A, vector of constants $\mathbf{b}$ *and the augmented matrix.*

(b) Solve the system of linear equations by applying row operations on the augmented matrix. State your steps clearly (example $2R_2 + R_1$).

Answer.

(a) $A = \begin{bmatrix} 1 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 & 5 \\ 2 & 2 & 0 & 1 & 5 \\ -3 & -3 & 1 & 3 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 3 \\ 4 \\ 9 \\ 1 \end{bmatrix},$

the augmented matrix $= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 2 & 3 \\ 0 & 0 & 1 & 1 & 5 & 4 \\ 2 & 2 & 0 & 1 & 5 & 9 \\ -3 & -3 & 1 & 3 & 1 & 1 \end{array}\right]$

(b)

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 2 & 3 \\ 0 & 0 & 1 & 1 & 5 & 4 \\ 2 & 2 & 0 & 1 & 5 & 9 \\ -3 & -3 & 1 & 3 & 1 & 1 \end{array}\right]$$

$$\xrightarrow[3R_1+R_4]{-2R_1+R_3} \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 2 & 3 \\ 0 & 0 & 1 & 1 & 5 & 4 \\ 0 & 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 3 & 7 & 10 \end{array}\right]$$

$$\xrightarrow[-R_2+R_4]{-2R_3+R_4} \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 2 & 3 \\ 0 & 0 & 1 & 1 & 5 & 4 \\ 0 & 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{-R_3+R_2} \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 2 & 3 \\ 0 & 0 & 1 & 0 & 4 & 1 \\ 0 & 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The above matrix corresponds to the following linear system:

$$\begin{aligned} x_1 + x_2 + 2x_5 &= 3 \\ x_3 + 4x_5 &= 1 \\ x_4 + x_5 &= 3 \end{aligned}$$

We can express x_1, x_3, x_4 in terms of x_2, x_5 :

$x_1 = 3 - x_2 - 2x_5$

$x_3 = 1 - 4x_5$

$x_4 = 3 - x_5$

Therefore, the solution set is

$$\left\{ \left[\begin{array}{c} 3 - x_2 - 2x_5 \\ x_2 \\ 1 - 4x_5 \\ 3 - x_5 \\ x_5 \end{array} \right] \middle| \; x_2, x_5 \text{ real numbers} \right\}$$

Exercise

Consider the following system of linear equations:

$$(*): \begin{cases} ax + y + bz = 1 \\ x + ay + bz = 1 \\ x + y + abz = b \end{cases}$$

where a, b are some real numbers.

(a) Solve the system $(*)$ when $a \neq -2, a \neq 1$ and $b \neq 0$.

(b) For each of the following cases, determine the value(s) of b so that (*) has solutions; solve the system when it has solutions.

i. $a = 1$,

ii. $a = -2$.

(c) What can you say about the system $(*)$ when $b = 0$?

Solution

(a) We solve by backward substitution. Consider the augmented matrix for the system:

$$\begin{bmatrix} a & 1 & b & 1 \\ 1 & a & b & 1 \\ 1 & 1 & ab & b \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 1 & ab & b \\ 1 & a & b & 1 \\ a & 1 & b & 1 \end{bmatrix}$$

$$\xrightarrow[-aR_1+R_3]{-1R_1+R_2} \cdot \begin{bmatrix} 1 & 1 & ab & b \\ 0 & a-1 & b-ab & 1-b \\ 0 & 1-a & b-a^2b & 1-ab \end{bmatrix}$$

$$\xrightarrow{R_2+R_3} \begin{bmatrix} 1 & 1 & ab & b \\ 0 & a-1 & b-ab & 1-b \\ 0 & 0 & 2b-ab-a^2b & 2-b-ab \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & ab & b \\ 0 & a-1 & b(1-a) & 1-b \\ 0 & 0 & b(1-a)(a+2) & 2-b-ab \end{bmatrix}$$

From the third row,

$$b(1-a)(a+2)z = 2-b-ab$$

Since $a \neq -2, a \neq 1$ and $b \neq 0$,

$$z = \frac{2-b-ab}{b(1-a)(a+2)} = \frac{-2+b+ab}{b(a-1)(a+2)}$$

From the second row,

$$(a-1)y + b(1-a)z = 1-b$$

Since $a \neq 1$,

$$y = \frac{1-b-b(1-a)z}{(a-1)} = \frac{a-b}{(a-1)(2+a)}$$

From the first row,

$$x + y + abz = b$$

$$x = b - y - abz = \frac{a-b}{(a-1)(2+a)}$$

(b) i. For $a = 1$, consider the augmented matrix for the system:

$$\begin{bmatrix} 1 & 1 & b & 1 \\ 1 & 1 & b & 1 \\ 1 & 1 & b & b \end{bmatrix} \xrightarrow{-R_2+R_3} \begin{bmatrix} 1 & 1 & b & 1 \\ 1 & 1 & b & 1 \\ 0 & 0 & 0 & b-1 \end{bmatrix}$$

The system is has solutions if and only if $b - 1 = 0$, i.e., $b = 1$.

The solution set in this case is $\left\{ \begin{bmatrix} 1-y-z \\ y \\ z \end{bmatrix} \middle| \, y, z \in \mathbb{R} \right\}$.

ii. For $a = -2$, consider the augmented matrix for the system:

$$\begin{bmatrix} -2 & 1 & b & 1 \\ 1 & -2 & b & 1 \\ 1 & 1 & -2b & b \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 1 & -2b & b \\ 1 & -2 & b & 1 \\ -2 & 1 & b & 1 \end{bmatrix}$$

$$\xrightarrow[2R_1+R_3]{-R_1+R_2} \begin{bmatrix} 1 & 1 & -2b & b \\ 0 & -3 & 3b & 1-b \\ 0 & 3 & -3b & 2b+1 \end{bmatrix} \xrightarrow{R_2+R_3} \begin{bmatrix} 1 & 1 & -2b & b \\ 0 & -3 & 3b & 1-b \\ 0 & 0 & 0 & b+2 \end{bmatrix}$$

The system is has solution if and only if $b + 2 = 0$, i.e., $b = -2$.

For $b = -2$, from the second row, we obtain :

$$-3y - 6z = 3$$
$$y = -1 - 2z$$

From the first row, we obtain :

$$x + y + 4z = -2$$
$$x = -2 - 4z - y = -1 - 2z$$

Thus the solution set in this case is $\left\{ \left[\begin{array}{c} -1-2z \\ -1-2z \\ z \end{array} \right] \middle| \, z \in \mathbb{R} \right\}$.

(c) When $b = 0$, the augmented matrix for the system becomes :

$$\left[\begin{array}{cccc} a & 1 & 0 & 1 \\ 1 & a & 0 & 1 \\ 1 & 1 & 0 & 0 \end{array} \right] \xrightarrow{R_1+R_2} \left[\begin{array}{cccc} a & 1 & 0 & 1 \\ a+1 & a+1 & 0 & 2 \\ 1 & 1 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{-(a+1)R_3+R_2} \begin{bmatrix} a & 1 & 0 & 1 \\ 0 & 0 & 0 & 2 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

From the second row, the corresponding equation is $0 = 2$, thus the system has no solution.

Exercise

Let A, B, C be matrices of the same size. The are related by the

$$A \xrightarrow{-2R_2+R_3} \xrightarrow{R_1 \leftrightarrow R_2} \xrightarrow{3R_1} B,$$

$$A \xrightarrow{-2R_2} \xrightarrow{3R_2+R_1} C,$$

Find a sequence of row operations transforming B to C.

Solution:

We reverse the row operations transforming A to B.

$$B \xrightarrow{-\frac{1}{3}R_1} \xrightarrow{R_1 \leftrightarrow R_2} \xrightarrow{2R_2+R_3} A$$

Combining the row operations transforming B to A, and then A to C, we obtain

$$B \xrightarrow{-\frac{1}{3}R_1} \xrightarrow{R_1 \leftrightarrow R_2} \xrightarrow{2R_2+R_3} \xrightarrow{-2R_2} \xrightarrow{3R_2+R_1} C$$

Exercise

Pove that RREF is unique.

Solution: Let A be a m*n matrix. we use induction on number of columns of A.
(i)For n=1, all 1 column matrix is either reduce to 0 vector or (1,0,...0)
(ii)Now suppose the statement is true for all k-column matrix.
For k+1-column matrix, by induction hypothesis, suppose we have to RREF say B and C for matrix A, then B and C are identical except maybe in last column.
Suppose $B \neq C$, then $B - C$ is a matrix that is all zero except some entries in last column.
Now let x be vector s.t. Bx=0, then Cx=0 (because row operation preserve solution) hence (B-C)x=0.
Expanded, $(B - C)\mathbf{x} = \mathbf{0}$ looks like

$$\begin{bmatrix} 0 & \cdots & 0 & d_1 \\ 0 & \cdots & 0 & d_2 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & d_m \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_{n+1} \end{bmatrix} = \begin{bmatrix} d_1 x_{n+1} \\ d_2 x_{n+1} \\ \vdots \\ d_m x_{n+1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Since we supposed $B \neq C$ with a difference in the last column on some row i, i.e. $d_i \neq 0$, we have $d_i x_{n+1} = 0, d_i \neq 0$, so x_{n+1} is forced to be 0.

Now the tricky sentence:
Thus x_{n+1} is not a free variable for either B or C
This is just restating the fact that x_{n+1} has been forced to 0.
So in each of these matrices the last column must contain a leading entry of 1 and have all the other entries 0(because that's what columns in RREF matrices corresponding to non-free variables look like)
By rules of RREF, we don't have any choice about where to put the 1 in the final column, so $B = C$ after all.

Quick Review (2nd tutorial) I

- How to compute the addition and scalar mutiplication of matrices?
- How to compute the product of two matrices?
- An insightful understanding of matrix multiplication.

Exercise

Compute the following. (a)

$$\begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 & -1 \\ 0 & 0 & 2 & 3 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 1 & 3 & 5 \\ -1 & 1 & 0 & -2 \\ 1 & 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}$$

(c)

$$\left(2\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 0 & 1 \\ -1 & -1 & 1 & 0 \end{bmatrix} - 3\begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 \\ 0 & -1 & -1 & -1 \end{bmatrix}\right) \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Solution:

(a) $\begin{bmatrix} 1 & 1 & 3 & 5 \\ -1 & -1 & 3 & 4 \\ 1 & 1 & 1 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 4 & 8 \\ -1 & 0 \\ 2 & 4 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 \\ -2 & -3 \\ -5 & -4 \end{bmatrix}$

Exercise

Let A, B be two square matrix of size n.
(a) Show that

$$(A+2B)(A-2B) = aA^2 + bAB + cBA + dB^2$$

for some real numbers a, b, c, d. Here $A^2 = AA$. Find a, b, c, d.
(b) Show that

$$(A+2B)(A-2B) = A^2 - 4B^2 \tag{1}$$

if and only if A, B commute, i.e. $AB = BA$.
(c) Let $n = 2$.
i. Find an example of A, B, such that all the entries of A, B are distinct and (1) is true.
ii. Find an example of A, B, such that all the entries of A, B are distinct and (1) is false.

Solution:
(a)

$$\begin{aligned}(A+2B)(A-2B) &= A(A-2B)+2B(A-2B)\\ &= A^2-2AB+2BA-4B^2\end{aligned}$$

Therefore $a=1, b=-2, c=2, d=-4$.
(b)

$$(A+2B)(A-2B) = A^2-2AB+2BA-4B^2 = A^2-4B^2$$

if and only if

$$A^2-2AB+2BA-4B^2-\left(A^2-4B^2\right) = \mathcal{O}$$

if and only if

$$-2AB+2BA = \mathcal{O}$$

if and only if

$$BA = AB$$

i. By part (b), if suffices to find A, B such that $AB = BA$.
For example, we can take $B = \alpha A$. Then

$$AB = A(\alpha A) = \alpha A^2 = (\alpha A)A = BA$$

Example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = 10A = \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix}$$

ii. By part (b), it suffices to find A, B such that $AB \neq BA$. For example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Exercise

1. *Let A, B, C be $3 * 3$ matrices. The are related by the following row operations.*

$$A \xrightarrow{-2R_2+R_3} \xrightarrow{R_1 \leftrightarrow R_2} \xrightarrow{3R_1} B,$$

$$A \xrightarrow{-2R_2} \xrightarrow{3R_2+R_1} C,$$

Find a matrix R s.t. $RB = C$.

Solution: recall in previous we have

$$B \xrightarrow{-\frac{1}{3}R_1} \xrightarrow{R_1 \leftrightarrow R_2} \xrightarrow{2R_2+R_3} \xrightarrow{-2R_2} \xrightarrow{3R_2+R_1} C$$

The corresponding operation matrix is:

$$\begin{bmatrix} -1/3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Remark

Remember to do the multiplication from right to left!!!

Answer: $\begin{bmatrix} 2 & 1 & 0 \\ 2/3 & 0 & 0 \\ -2/3 & 0 & 1 \end{bmatrix}$